

DNQ-DLC: Dynamic-Neighbourhood Quality-Guided Deep Latent Competition for Closed-Loop Multi-Vehicle Overtaking in Autonomous Racing

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Abstract—Racing overtakes are decided within seconds, and the constraining opponent can change in that window. DNQ-DLC is a receding-horizon planner with three parts: a dynamic neighbourhood that rebuilds the ego-centric interaction graph from each observation and decides which opponents enter it, an action-conditioned world model that rolls candidate ego actions into imagined futures, and a quality-guided score over progress, containment, interaction and stability that ranks those futures. Over 48 matched cases per method, five draws of the DNQ-DLC recipe complete 45.8 of 48 cases on average, against 41.0, 41.5 and 45.5 for three world-model variants retrained on the same data, 45 for a rule expert and at most 35.8 for three continuous-control families, and reach the strict tier (sustained lead, no contact, on track, stable afterwards) in 20.2 cases against 17.5, 18.5 and 13.5 for the three variants, 12 for the rule expert and at most 6.4 for the continuous-control families. The strict-tier range overlaps every variant, while the rank-gain and containment ranges are disjoint from every comparator draw. Every draw gains more positions per case, 4.62 to 5.02, and spends 12.4–19.9 % of the episode off the racing surface against 25.0–49.7 % for every variant draw. The admitted set saturates at its three-slot budget, the imagined ego position stays within 1.2 m over four steps, the planner leaves its rule anchor in 92–93 % of decisions, and the shield is active on 30–46 % of steps. The claims are limited to the tested simulator and vehicle counts.

Index Terms—Autonomous racing, overtaking, world models, dynamic interaction graphs, multi-agent decision making, receding-horizon planning.

I. INTRODUCTION

MULTI-VEHICLE overtaking in autonomous racing is decided within a short window. An opportunity appears when the vehicle ahead is slower at corner exit or leaves a line open, and it closes within a few seconds, so the decisive part of the manoeuvre is a small share of any longer episode, and rare interactions of this kind drive test generation for automated driving [1], [2]. The opponent that constrains the manoeuvre also changes during the pass: the vehicle that sets the initial closing strategy gives way to an alongside vehicle, or to a vehicle that constrains the pass from a different line. A controller therefore has to rebuild its interaction context from each new observation, estimate what a candidate action does over the next few tenths of a second, and trade progress against track retention, contact risk and stability. That trade-off has no free setting: an aggressive action can secure position while removing lateral clearance, and a conservative one can preserve containment while the passing window closes [3],

[4]. Whatever the controller chooses, it has to remain inside the capability of the vehicle at racing speed [5]–[8].

The relevant-opponent set is not a fixed list. A leading vehicle may set the initial closing strategy, an adjacent vehicle may define the corridor that is open, and a vehicle that was far away can become the binding constraint after it brakes or abandons a pass [9], [10]. A controller that keeps a fixed set of interaction objects, or assigns roles once and reuses them, will keep reasoning about an opponent that no longer matters while ignoring the vehicle that does.

Four lines of work bear on this problem. Dynamics-aware racing planners enforce track and vehicle limits, but usually over predefined opponents or pairwise roles [11], [12]. Learned policies produce nonlinear passing behaviour, at the cost of sensitivity to observation layout, reward construction and the traffic distribution seen in training [13]–[16]. World-model planners compare actions through imagined futures, yet a rollout that predicts progress well says nothing about whether the pass stays on the track or enters an uncertain state [17], [18]. Variable-agent models do represent changing traffic sets, but are mostly built for motion forecasting or scene simulation rather than for closed-loop racing decisions [19]–[21].

We propose Dynamic-Neighbourhood Quality-Guided Deep Latent Competition (DNQ-DLC). Its dynamic neighbourhood decides which opponents enter the interaction representation, and is rebuilt from every real observation rather than held fixed, so the selected set can change while the manoeuvre is under way. Its deep latent competition is an action-conditioned graph world model that rolls candidate ego actions into short-horizon imagined futures, and its quality guidance ranks those futures by progress plus explicit track containment, interaction quality, boundary-risk and variance terms. Only the first action of the selected candidate is executed.

The main contributions are:

- We treat the relevant-opponent set as a decision of the closed-loop problem rather than as a fixed input. Relevance is re-evaluated after each real observation and resolved by the planner itself, so the interaction representation follows the binding constraint instead of a preassigned role or a predetermined pair.
- We condition the prediction on the ego action, so competing manoeuvres are compared on their imagined short-horizon futures rather than on a single-step estimate.

- We score those futures by the quality of the overtake they produce rather than by predicted progress alone, adding track containment, manoeuvre quality and stability to the ranking.
- We evaluate the complete controller through matched simulator comparisons across four field sizes, with state-level and controller-internal analyses read from those same runs and diagnostic failure cases. The conclusions are limited to the tested simulation conditions.

Section II positions the work against overtaking control, world-model planning and variable-agent interaction modelling; Section III presents DNQ-DLC; Section IV evaluates overtaking completion and its trade-offs; Section V concludes the paper.

II. RELATED WORK

A. Overtaking Planning and Control

Autonomous-racing planners combine opponent prediction, collision margins, track constraints and dynamically feasible motion generation [3], [4]; vehicle-limit planning connects the resulting trajectories to the available chassis capability [7], [8]; and competitive racing systems put both to work [11], [12], [22], [23]; Safety is enforced elsewhere by a reference-governor layer [24], by game-theoretic decision making against a predicted opponent [25], and under command-to-actuation delay [26]. In many of those formulations, though, the interaction is defined around a fixed lead vehicle, a predetermined pair, or opponent roles assigned once, an assumption that breaks down when braking, overlap or a change of order makes another vehicle decisive.

Learning-based policies reduce the dependence on explicit manoeuvre rules. Reinforcement learning and motion-image representations have been used for competitive driving, overtaking and lane-change decisions, the last of these coupling scene evolution to manoeuvre selection [13], [27], [28]. Integrated stacks learn perception and planning representations jointly [29]–[31]. Their behaviour stays tied to observation semantics, reward design and training coverage [14], [15]: a policy trained with a fixed vehicle ordering loses its meaning when another opponent becomes the one that limits the manoeuvre.

B. World-Model Planning

World models compare actions before they are executed. Latent and generative models have made imagined rollouts practical for control and driving [18], [32], [33], risk-aware variants price the uncertainty of the imagined outcome [34], and Deep Latent Competition introduced action-conditioned latent prediction for racing, which is also the basis of the DLC-JTO reference used in this study [17]. Recent driving world models address distribution change during planning [35]–[37], and vision-language-action world models condition the prediction on language [38].

For overtaking, however, an accurate progress forecast is not enough. A rollout can predict the along-track gain correctly and still leave the track, force a contact, or commit to a state

the model has never seen, and a model whose opponent slots are fixed carries an outdated interaction context through the whole horizon. The step this paper addresses is therefore the combination of a short-horizon prediction with the opponents that matter at the current real decision, evaluated by the quality of the resulting overtake rather than by progress alone. Cost and constraint design in the optimising planners is typically specific to the track and the vehicle limits [7], [11], [12]; the geometry and feasibility terms here are of that kind, while the manoeuvre-quality term is learned from expert passes.

C. Variable-Agent Interaction Representations

Motion-forecasting datasets and models accommodate variable numbers of interacting agents directly [39], [40]. Scene Transformer and Wayformer exchange information across agents with joint attention [19], [41]; HiVT, Motion Transformer and QCNet organize interaction hierarchically, by intention, or around queries [20], [21], [42]; a dynamic graph has also been learned for the overtaking manoeuvre itself [43]. Planning approaches connect multimodal goals or vectorized scene tokens to motion generation [44], [45], graph encoders have been coupled to model-predictive control for interactive decision making [46], and traffic simulation broadens the range of interactive scenes that can be evaluated [10], [47], [48].

These methods establish that a changing agent set can be encoded, but most of them then optimize forecasting or scene generation, so the representation is rarely tested as an active part of a closed-loop overtake, where the selected opponent has to change what gets predicted and what gets executed. Open-loop prediction quality is also known not to transfer into planning utility by itself [49], [50].

D. Research Gap and Positioning

This paper asks whether relevance-based opponent selection and manoeuvre-quality scoring improve a world-model overtaking controller when the traffic configuration changes. DNQ-DLC combines masked interaction encoding and receding-horizon planning, both established, with an explicit relevance selector and a quality-weighted candidate score; the claim is about that combination, not about attention, graph encoders or model-predictive selection taken separately. A dynamic graph has been learned for overtaking [43]; the question here is what the selected set does inside a transition model that is itself under test.

Because the candidate ranking depends on an imagined future, the learned transition is part of the object under test rather than an assumed-accurate component.

III. METHODOLOGY

DNQ-DLC performs receding-horizon control over an ego-centric interaction graph. Selection changes which opponents are represented without changing the width of the learned interface, so one checkpoint serves every traffic configuration. The subsections below define the inputs, the representation, the transition model and the decision rule, together with the recovery branch that prices imagined states outside the training

support, supplies a return command when the real state has already left it, and takes over only when no candidate avoids the guarded region.

A. Problem Formulation and Closed-Loop Framework

Fig. 1 organizes the controller into three stages, executed once per real decision step. The first encodes the dynamic neighbourhood: the ego-relative relations of the exposed vehicles are ranked and the retained ones packed into fixed slots with validity masks (Section III-B). The second predicts the next state, reward, boundary-risk proxy and log-variance over the imagined horizon from the self, relation and action encoders (Section III-C). The third rolls a candidate pool through that same model, scores the rollouts and commits the first action of the best candidate (Section III-D). The figure labels the slot block by a single count because the three capacities coincide at the submitted operating point.

Let $\mathcal{M}$ denote the track map and let $X_t = [x_t^i]_{i=1}^N$ be the joint telemetry at physical time t . The controlled vehicle is indexed by e ; all other rows are background vehicles. The normalized control action is

$$a_t^e = [\delta_t, u_t, b_t]^\top \in \mathcal{A}, \quad (1)$$

where steering, throttle and braking are bounded by the simulator action space. The planner maintains a candidate set $\mathcal{C}_t$ and an imagined horizon H . For candidate n , J_n denotes the discounted predicted quality of its continuation. The receding-horizon decision is

$$n^* = \arg \max_{1 \leq n \leq |\mathcal{C}_t|} J_n, \quad a_t^e = a_t^{(n^*)^e}. \quad (2)$$

Only a_t^e is committed to the simulator. The next telemetry X_{t+1} starts a new selection–packing–prediction cycle, so an imagined prediction never changes the physical graph by itself.

B. Dynamic Ego-Centric Interaction Graph

For each vehicle j exposed by the runtime telemetry builder as a candidate relation, the extractor forms an ego-relative relation

$$r_t^{ej} = \begin{bmatrix} \Delta f_{b,t}^{ej}, \Delta l_{b,t}^{ej}, \Delta v_{x,t}^{ej}, \Delta v_{y,t}^{ej}, d_t^{ej}, \\ \sin \Delta \psi_t^{ej}, \cos \Delta \psi_t^{ej} \end{bmatrix}^\top \in \mathbb{R}^{d_r}, \quad (3)$$

where Δf_b and Δl_b are displacements in the heading-defined telemetry frame, Δv_x and Δv_y are global-frame relative velocity components, d is separation, and $\Delta \psi$ is heading difference. Displacements and distance are divided by the simulator spatial scale; velocities are divided by its speed scale before encoding. The relation channels follow the simulator's physical forward and left vehicle axes, so Δl_b is a true cross-track offset and $\Delta \psi$ the heading error against the local track tangent. The channels are read in those vehicle axes rather than projected onto the tangent, which would not measure a cross-track distance.

Let $\mathcal{V}_t^{\text{cand}}$ denote the bounded candidate set exposed by the telemetry builder at the current physical step. A candidate is eligible if it lies inside radius R_{loc} or satisfies a forward,

alongside, or rear-pressure gate. With gate indicators I_f, I_s, I_r , this gives

$$\mathcal{N}_t = \{j \in \mathcal{V}_t^{\text{cand}} : d_t^{ej} \leq R_{\text{loc}} \vee I_f^j \vee I_s^j \vee I_r^j\}. \quad (4)$$

All quantities in the gate and score are converted back to metric units. For each eligible vehicle the selector computes

$$\begin{aligned} \rho_t^j = & w_f G_f(x_t^e, x_t^j) + w_s G_s(x_t^e, x_t^j) \\ & + w_r G_r(x_t^e, x_t^j) + w_d q_d + w_{v,\rho} q_v \\ & + w_{\psi,\rho} q_\psi - w_{d,\rho} d_t^{ej}, \end{aligned} \quad (5)$$

where I_f tests $0 < \Delta f_b < f_f, |\Delta l_b| < l_f$; I_s tests $|\Delta f_b| < f_s, |\Delta l_b| < l_s$; and I_r tests $-f_r < \Delta f_b \leq 0, |\Delta l_b| < l_r, q_v > v_r$. The continuous factors are

$$\begin{aligned} G_f &= I_f[1 - |\Delta f_b - f_0|/f_w]_+, \\ G_s &= I_s[1 - |\Delta l_b|/l_s]_+, \quad G_r = I_r, \\ q_d &= 1/\max(d/d_0, 1), \\ q_\psi &= [\cos \Delta \psi]_+ - c_h |\sin \Delta \psi|. \end{aligned} \quad (6)$$

Here $[z]_+ = \max(z, 0)$, and q_v is the closing speed obtained by projecting the global-frame relative velocity on the controlled vehicle's own heading axis, so it does not depend on the orientation of the world axes. Numerical thresholds and weights of (5) are collected in Table I. The runtime sorts the represented neighbours and returns an ordered, capacity-limited list

$$\begin{aligned} (j_t^1, \dots, j_t^{K_t}) &= \text{Select}_{K_{\text{cap}}}(\mathcal{N}_t; \rho_t), \\ \tilde{\mathcal{N}}_t &= \{j_t^s\}_{s=1}^{K_t}, \quad K_t = |\tilde{\mathcal{N}}_t| \leq K_{\text{cap}}. \end{aligned} \quad (7)$$

The list order is the slot order: for the controlled row e , the represented neighbour j_t^s occupies relation slot s . Both $\mathcal{N}_t$ and this slot correspondence can change at every real decision time. K_{cap} is a computational capacity, not a fixed physical neighbourhood. We use K_{model} for the fixed relation-slot width expected by the trained checkpoint and decoder; it may be smaller than the runtime capacity after repacking.

Each selected relation is paired with a validity indicator $m_t^{i,s} \in \{0, 1\}$. Because the learned checkpoint has a fixed interface, only the first $\tilde{K}_t = \min(K_t, K_{\text{model}})$ selected relations are packed into that interface. For the controlled row, model slot s contains $(r_t^{ej_t^s}, 1)$ when $s \leq \tilde{K}_t$ and a masked fill slot otherwise. The remaining ranked relations are retained only in the runtime selection record and cannot enter the fixed-width decoder tensor. The same construction applies to other model rows when their local relations are represented. The resulting slot vector is

$$y_t^{i,s} = [r_t^{i,s}, m_t^{i,s}]^\top \in \mathbb{R}^{d_y}, \quad d_y = d_r + d_m, \quad d_m = 1, \quad (8)$$

with $m = 0$ for padded slots; the mask suppresses their physical fill values. The packed row observation is

$$\begin{aligned} \bar{o}_t^i &= [\phi_t^i, y_t^{i,1}, \dots, y_t^{i,K_{\text{model}}}]^\top \in \mathbb{R}^{d_o}, \\ d_o &= d_\phi + K_{\text{model}} d_y. \end{aligned} \quad (9)$$

where ϕ_t^i contains the self state and local track descriptors. Three distinct quantities are therefore involved and are kept apart throughout: K_{cap} is the runtime capacity at which

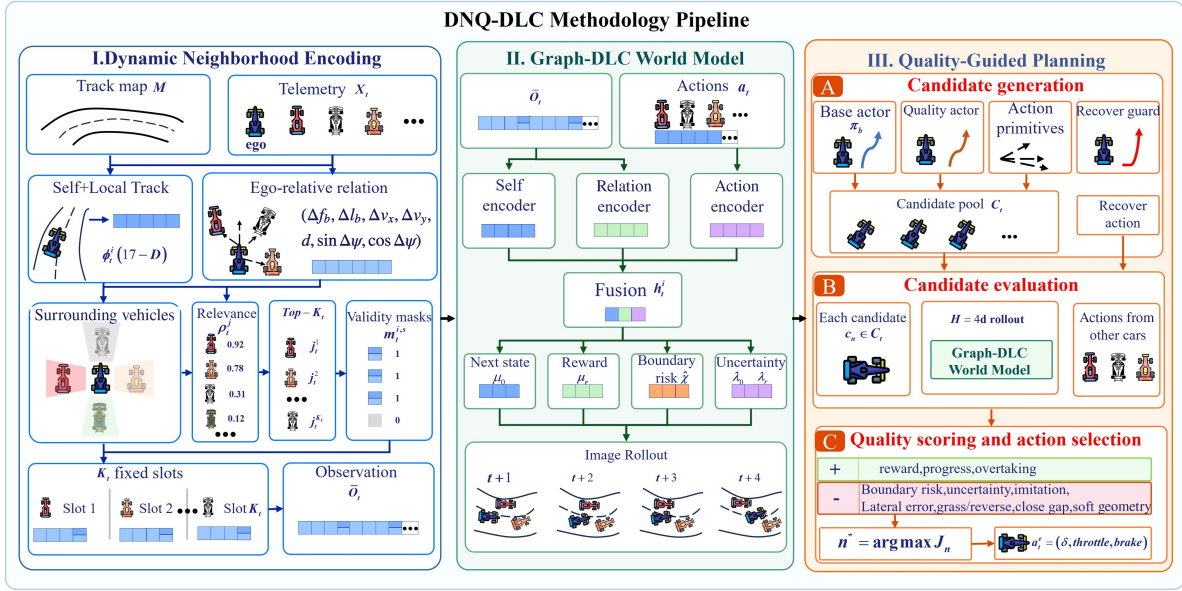


Fig. 1. Overview of DNQ-DLC. Stage I selects the dynamic neighbourhood and packs the retained relations into masked slots K_t ; Stage II predicts the next state, reward, boundary-risk proxy and log-variance of each candidate action; Stage III scores the imagined rollouts and commits the first action of the best candidate unless the recovery guard fires.

opponents are ranked, K_t is how many real relations were selected at this step, and K_{model} is the width the checkpoint was trained with. The graph is an ego-centred star. Selected vehicles are nodes, their ego-relative features are the incident edges, and the learned interaction operator is a masked mean over those edges.

C. Action-Conditioned Graph World Model

For agent row i , self, relation and action encoders are

$$e_s^i = f_s(\phi_t^i), \quad e_r^{i,s} = f_r(r_t^{i,s}), \quad e_a^i = f_a(a_t^i). \quad (10)$$

Masked aggregation prevents padded relations from contributing:

$$e_r^i = \frac{\sum_{s=1}^{K_{\text{model}}} m_t^{i,s} e_r^{i,s}}{\max(\sum_{s=1}^{K_{\text{model}}} m_t^{i,s}, 1)}. \quad (11)$$

The local and scene-level summaries are fused as

$$h_t^i = f_h[e_s^i, e_r^{i,s}, e_a^i, \bar{e}_s, \bar{e}_a], \quad (12)$$

$$\bar{e}_s = \text{mean}_i e_s^i, \quad \bar{e}_a = \text{mean}_i e_a^i.$$

The transition heads output the next packed observation, reward, boundary-risk logit and log-variance outputs:

$$(\mu_{t+1|t}^{o,i}, \lambda_{t+1|t}^{o,i}, \mu_{t+1|t}^{r,i}, \lambda_{t+1|t}^{r,i}, z_{t+1|t}^{\chi,i}) = F(h_t^i), \quad (13)$$

$$\hat{\chi}_{t+1|t}^i = \sigma(z_{t+1|t}^{\chi,i}).$$

The planner uses the model's predicted means. Training minimizes the weighted point-prediction objective

$$\begin{aligned} \mathcal{L}_o &= \text{MSE}(\mu_{t+1|t}^o, \bar{o}_{t+1}), \\ \mathcal{L}_r &= \text{MSE}(\mu_{t+1|t}^r, r_{t+1}), \\ \mathcal{L}_\chi &= \text{BCEWithLogits}(z_{t+1|t}^\chi, \chi_{t+1}), \\ \mathcal{L}_{\text{prop}} &= \mathbb{E}_i \text{MSE}(\pi_b(\bar{o}_t^i), a_t^i), \\ \mathcal{L} &= \mathcal{L}_o + \alpha_r \mathcal{L}_r + \alpha_\chi \mathcal{L}_\chi \\ &\quad + \alpha_p \mathcal{L}_{\text{prop}} + \lambda_q \mathcal{L}_q + \mathcal{L}_v, \\ \mathcal{L}_v &= 0.01 (|\mathbb{E} \lambda^o| + |\mathbb{E} \lambda^r|). \end{aligned} \quad (14)$$

Here π_b is the base actor and $\mathcal{L}_q$ is an optional auxiliary quality-head loss. In the submitted configuration $\lambda_q = 0$; quality guidance is supplied by a separately behaviour-cloned quality actor π_q , not by an active auxiliary prediction head. MSE averages squared scalar errors over samples and output entries; base-actor supervision includes all vehicle rows. BCE and variance expectations use mean reductions over the batch and represented rows. Here r_{t+1} is the reward returned after executing a_t . In the rollout equations below, $\hat{r}_{t+\tau}^{(n)}$ denotes the predicted reward for the transition from $t + \tau$ to $t + \tau + 1$, indexed by its starting time; at $\tau = 0$ it is the ego-row reward mean $\mu_{t+1|t}^{r,e}$. The variance outputs receive this regularizer but no likelihood-based calibration loss, so C_u is a proxy for how far the imagined transition sits from the fitted mean, not a calibrated predictive variance, and it is used at the small weight the table gives it: over 7,920 transitions sampled from the released collection episodes the two predicted log-variances sum to a channel with standard deviation 0.04 that never leaves $[-0.5, 0.5]$, so its contribution to J_n is about four per cent of the risk term's. The risk target is a telemetry flag for grass, backward motion, or excessive absolute lateral telemetry; it does not estimate collision probability.

The world model and base actor use 160 collection episodes with 352,000 transitions from expert telemetry policies and 4–6 vehicles. In every episode the recorded vehicle is driven by the joint-clearance pilot and the remaining vehicles hold a constant 13 m/s lane target. Adam training runs for 40,000 updates of batch size 512 with a learning rate decaying from 3×10^{-4} on a cosine schedule. The quality actor minimizes $\mathcal{L}_{\text{BC}} = \mathbb{E}_{(o,a) \sim \mathcal{D}_q} \text{MSE}(\pi_q(o), a)$ on 1,526 accepted on-track event samples from 24 separate collection episodes. Failed events are excluded from $\mathcal{D}_q$; this is a conditional imitation prior, not learned evidence of safe behaviour in failure states. Episode-disjoint predictive validation is performed after freezing the checkpoint and is not used for model selection:

the submitted transition model is the endpoint of the fixed 40,000-update cosine schedule, so no validation quantity was consulted before that endpoint was fixed. Section IV reports the realized error of that imagined rollout.

Fusing the draws: The recipe above is stochastic, and Section IV reports five draws of it whose strict-tier counts span fourteen to twenty-six of the forty-eight cases; the submitted controller is the first of them. Because retraining the transition model moves the endpoint counts by more than the gaps between the controllers compared below, Section IV reports every learned method as the mean over the draws of its recipe rather than as a single checkpoint, and the fused variant defined below is released as an alternative to the submitted controller. The transition model of (13) is trained $M = 5$ times on the same data under the same schedule, and the planner consumes the fused heads

$$\begin{aligned}\bar{\mu}_{t+1|t}^o &= \frac{1}{M} \sum_{j=1}^M \mu_{t+1|t}^{o,(j)}, \\ \bar{\lambda}_{t+1|t}^o &= \log\left(\frac{1}{M} \sum_{j=1}^M e^{\lambda_{t+1|t}^{o,(j)}} + \text{Var}_j[\mu_{t+1|t}^{o,(j)}]\right),\end{aligned}\quad (15)$$

where the superscript (j) indexes the member. The first term inside the logarithm is the aleatoric variance the members report between them, and the second is their disagreement, which no member can see on its own [51], [52]. Equation (25) reads the uncertainty cost off the same log-variance, so a candidate whose outcome the members disagree about is penalized in proportion to that disagreement. Reward, risk logit and the auxiliary quality head are averaged in the same way. The proposal actor, the quality actor and the planner constants are not fused: they are single instances shared by the members, so (15) is the only difference between the fused variant and the controller of Table II. Fusion multiplies the cost of one imagined rollout by M and raises the mean decision time from 243 ms to 443 ms over the 48 cases; it is released with the controller and is not the reported result. No evaluation case was used to select the members or to weight them. Both are wall-clock means, read against the control interval: the simulator advances 0.02 s per step, so the four-step ranking spans 0.08 s while one decision costs 243 ms. Neither the single model nor the fused policy ranks inside one control interval on the evaluation machine, and the evaluation applies each decision at the next step without simulating that delay, which is itself a control problem in racing [26], so the endpoints below describe the algorithm rather than a real-time control loop. Closing the gap would take a smaller candidate budget, a shorter horizon or a batched rollout; the fused policy would also need fewer members.

D. Quality-Guided Predictive Planning

The candidate set combines learned proposals and interpretable action primitives:

$$\begin{aligned}\mathcal{U}_t &= \left\{ \pi_b(\bar{o}_t^e), \pi_q(\bar{o}_t^e), c_{\text{chase}}, c_{\text{anchor}}, \right. \\ &\quad \left. c_{\text{pass}}, c_{\text{brake}}, c_{\text{rec}}, c_{\text{ret}}, c_{\text{pert}} \right\}, \\ \mathcal{C}_t &= \text{Unique}_B(\mathcal{U}_t).\end{aligned}\quad (16)$$

Unique_B removes duplicate normalized actions and retains at most the configured candidate budget B . In the pool, π_b and π_q are the learned proposal and quality actors; c_{anchor} is the command of a rule-based telemetry expert gate of the same family as the rule-expert reference; c_{chase} closes up on the vehicle ahead; c_{pass} and c_{brake} are the lane-change and braking manoeuvres; c_{rec} and c_{ret} are the recovery action and its graded corridor-return variants, both defined below; and c_{pert} is a fixed perturbation profile around the proposal. Templates are enabled by their corresponding telemetry gates, and their constants belong to the released planner code.

A feasibility guard handles states in which ordinary overtaking candidates would be extrapolated outside the training support. Let ζ_t indicate backward motion, severe track departure, or an unsafe heading. While it fires, the guard's recovery command is one candidate among the others; the committed action is

$$\begin{aligned}\mathcal{E}_t &= \begin{cases} \{n : \phi^{(n)} = 0\}, & \zeta_t^{\text{esc}} = 1, \\ \{1, \dots, |\mathcal{C}_t|\}, & \text{otherwise,} \end{cases} \\ a_t^e &= \begin{cases} a_t^{\text{rec}}, & \zeta_t^{\text{esc}} = 1 \text{ and } \mathcal{E}_t = \emptyset, \\ a_t^{(n^*),e}, & \text{otherwise, } n^* = \arg \max_{n \in \mathcal{E}_t} \tilde{J}_n, \end{cases}\end{aligned}\quad (17)$$

where $\mathcal{E}_t$ is the eligible set and $\tilde{J}_n$ is the constrained score defined below. Here a_t^{rec} is the corridor-return action the reported runs execute. Its steering component is a bounded cross-track return law,

$$a_t^{\text{rec},s} = \text{clip}(g_r \varsigma_t - g_\psi \Delta\psi_t, -1, 1), \quad (18)$$

where ς_t is the signed cross-track channel of the corrected telemetry, $\Delta\psi_t$ is the heading error to the local track tangent in radians, and the two gains are $g_r = 1.4$ and $g_\psi = 1.6$. The sign of ς_t is fixed by the vehicle's own lateral response, measured once in the simulator, not read off the protocol: a vehicle placed off the centreline returns with $\text{sgn}(a_t^{\text{rec},s}) = \text{sgn}(\varsigma_t)$ and diverges with the opposite sign. Throttle is 0.55 below a 17.5 m/s reference that is reduced by the corner and lateral errors and floored at 12 m/s, and 0.08 above it, with braking applied 3 m/s above that reference. The command passes through the same speed and geometry limiter as the candidate actions, so it respects the same vehicle limits as everything else, and the candidate pool additionally carries the archived rule-recovery template when the real state is infeasible. This guard is enabled for the reported DNQ-DLC runs. It is a feasibility fallback rather than a learned policy. The escalation indicator ζ_t^{esc} is defined with the state-feasibility violation below, and the escalation window and the guard thresholds appear in Table I.

Equation (18) replaces the archived track-follower, which aimed at the centreline eight tiles ahead and folded the offset into a heading command, $1.05 \ell/40$, wrapped to $(-\pi, \pi]$. At that gain the lateral term reaches the steering lock only beyond about 25 m of offset, while the runs contain vehicles 10 to 17 m off the centreline that took hundreds of steps to return or never returned. Raising the gain is not a repair, because the wrap then enters the operating range: at a gain of 8 a 17 m offset inverts the command and drives the vehicle outward. Three properties of the return layer follow from the same

measurement. It is gated at 0.90 half-widths so a pass keeps its full lateral envelope inside that offset. It defers while a vehicle is alongside and the ego is still inside the road edge, because cutting back to the racing line with a car in the adjacent lane is a contact rather than a recovery; outside the road edge it acts unconditionally. When it acts, its graded variants c_{ret} are inserted at the head of $\mathcal{C}_t$ and it also overrides the steering component of a_t^{rec} . The layer was fixed before the 48-case evaluation was re-run, and its effect was checked on six dense-field episodes outside those 48 cases (seeds 33 to 38 at six vehicles, under the submitted world-model draw): the strict tier rises from two to three of the six, the in-corridor share of the manoeuvre window from 0.88 to 0.98, and the whole-episode grass fraction falls from 0.112 to 0.064, at a completion of six either way. Six episodes are too few to separate the two layers, so that check is reported as a direction rather than as an effect.

For each candidate, the first target action is fixed to the candidate and later target actions follow a quality-guided continuation

$$\begin{aligned} a_t^{(n),e} &= c_n, \\ \tilde{a}_{t+\tau}^{(n),e} &= (1 - \alpha_q) \pi_b(\hat{O}_{t+\tau|t}^{e,(n)}) \\ &\quad + \alpha_q \pi_q(\hat{O}_{t+\tau|t}^{e,(n)}), \\ a_{t+\tau}^{(n),e} &= \mathcal{S}_t(\tilde{a}_{t+\tau}^{(n),e}, \hat{O}_{t+\tau|t}^{e,(n)}), \quad \tau > 0. \end{aligned} \quad (19)$$

Here $\mathcal{S}_t$ is the deterministic speed/geometry limiter also applied before candidate deduplication: it caps throttle, increases braking, and attenuates steering when speed or geometry thresholds are exceeded. The quality policy includes its own relation repacking before inference. Non-target actions are generated by the configured background-policy set Π_{bg} . Let $\mathcal{F}$ denote the complete encoder–transition–decoder map whose output heads are F in (13). It is iterated as

$$\begin{aligned} (\hat{O}_{t+\tau+1|t}^{(n)}, \hat{r}_{t+\tau}^{(n)}, \hat{\chi}_{t+\tau+1|t}^{(n)}) &= \mathcal{F}(\hat{O}_{t+\tau|t}^{(n)}, a_{t+\tau}^{(n)}), \\ \hat{O}_{t|t}^{(n)} &= \bar{O}_t. \end{aligned} \quad (20)$$

The transition tensor preserves slot positions and validity masks throughout the imagined horizon. The quality-actor input is repacked from the predicted relations at each continuation step, which can reorder the represented slots but cannot introduce an unobserved vehicle; only the next real observation refreshes the physical candidate set. Because the selector re-ranks at every real step, a fixed slot position does not by itself guarantee that a decoder channel tracks the same physical vehicle across a rank change. Within one decision the correspondence is consistent, which is what the candidate scores rely on; across decisions the graphs are rebuilt from telemetry rather than carried over.

The candidate score is

$$\begin{aligned} J_n &= \sum_{\tau=0}^{H-1} \gamma^\tau \left[\hat{r}_{t+\tau}^{(n)} + \lambda_p R_p^{(n,\tau)} + \lambda_o R_o^{(n,\tau)} \right. \\ &\quad - \lambda_\chi C_\chi^{(n,\tau)} - \lambda_u C_u^{(n,\tau)} - \lambda_{\text{live}} C_{\text{live}}^{(n,\tau)} \\ &\quad - \lambda_i C_i^{(n,\tau)} - \lambda_\ell C_\ell^{(n,\tau)} - \lambda_g C_g^{(n,\tau)} \\ &\quad - \lambda_c C_c^{(n,\tau)} - \lambda_f C_f^{(n,\tau)} \\ &\quad \left. + \lambda_r R_r^{(n,\tau)} \right]. \end{aligned} \quad (21)$$

The progress, risk and stall terms are

$$\begin{aligned} R_p^{(n,\tau)} &= \hat{\eta}_{t+\tau+1|t}^{(n)} - \hat{\eta}_{t+\tau|t}^{(n)}, \quad C_\chi^{(n,\tau)} = \hat{\chi}_{t+\tau+1|t}^{(n)}, \\ C_{\text{live}}^{(n,\tau)} &= [v_{\text{live}} - \hat{v}_{t+\tau+1|t}^{(n)}] / v_{\text{live}}, \\ R_o^{(n,\tau)} &= w_{\text{pass}} |\{j \in \mathcal{A}^\tau : \tilde{\lambda}_j^{\tau+1} \leq -\ell_{\text{clr}}\}| \\ &\quad + \mathbb{I}[\mathcal{A}^\tau \neq \emptyset, \mathcal{A}^{\tau+1} \neq \emptyset] \\ &\quad \times \text{clip} \left(\frac{\min_j \tilde{\lambda}_j^\tau - \min_j \tilde{\lambda}_j^{\tau+1}}{L_g}, -r_-, r_+ \right) \\ &\quad + w_a [|\mathcal{A}^\tau| - |\mathcal{A}^{\tau+1}|]_+ + w_v [v_{t+\tau+1} - v_c]_+. \end{aligned} \quad (22)$$

Here η is normalized visited-track progress, v_{live} is the pace below which a rollout is priced as stalled, and

$$\begin{aligned} \tilde{\lambda}_j &= \rho_j^\parallel \cos e_\psi + \rho_j^\perp \sin e_\psi, \quad \tilde{\lambda}_j = \rho_j^\parallel \sin e_\psi - \rho_j^\perp \cos e_\psi, \\ \mathcal{A}^\tau &= \{j : 0 < \tilde{\lambda}_j^\tau < L_c, |\tilde{\lambda}_j^\tau| < L_\perp\}. \end{aligned} \quad (24)$$

$\rho_j^\parallel, \rho_j^\perp$ are the packed body-frame slot relations of neighbour j and e_ψ is the target's own heading error, so (24) is the same relation expressed in the frame of the local track tangent; $\mathcal{A}^\tau$ is the set of vehicles still ahead of the ego inside the approach cone and $|\mathcal{A}^\tau|$ is n_{ahead} . The cone has length L_c and lateral half-width $L_\perp$, both in metres; $L_\perp$ is wider than the simulated road, so the cone acts as a longitudinal gate rather than a lateral one. L_g appears only in the denominator of the graded term of (23) and is the length that maps a lead change onto the clip $[-r_-, r_+]$; it is set equal to L_c , so closing one cone length saturates the grading. The same quantity enters the overtaking credit: an opponent counts as cleared in (23) once its track-frame lead $\tilde{\lambda}_j$ passes $-\ell_{\text{clr}}$, and the lead change over one step is measured as the change in the smallest λ_j over the approach cone. The numerical values of w_{pass} , ℓ_{clr} , L_g , $r_\pm$, w_a , w_v and v_c are in Table I. The frame matters because a body-frame cone can be left by yaw alone: with the body-frame version of this term, a candidate that steered hard pushed the leading vehicle out of the cone and collected the full clearance credit without changing the along-track order, so the planner could hold a hard steering angle while the order ahead stayed the same. The track-frame lead $\tilde{\lambda}_j$ in (24) changes only when the along-track order changes, so the credit has to be earned by closing up or by passing. The overtaking reward is replaced by $\min(R_o, 0)$ when the predicted target is on grass, moves backward, or satisfies $|\hat{\ell}| > \ell_g$. The remaining costs use the predicted target's lateral telemetry $\hat{\ell}$ and heading deviation $e_\psi = \arccos(\text{clip}(c_\psi, -1, 1))$. The proposal set is $\Pi = \{\pi_b, \pi_q\}$, with $\tilde{a}_{\pi,t} = \pi(\hat{o}_t^e)$. These imitation references

are held at their real-step values even though continuation actions are recomputed.

$$\begin{aligned}
C_\ell &= |\hat{\ell}| + w_\psi |e_\psi| + w_{\ell,e} [|\hat{\ell}| - \ell_e]_+ \\
&\quad + w_{\psi,e} [c_{\psi,\min} - \cos(e_\psi)]_+, \\
C_g &= w_{\text{grass}} \mathbb{I}[\text{grass}] + w_{\text{back}} \mathbb{I}[\text{back}] \\
&\quad + w_{\ell,g} [|\hat{\ell}| - \ell_g]_+, \\
C_c &= [g_c - d_{\text{front}}]_+ / g_c, \\
C_u &= \text{clip}(\lambda^o + \lambda^r, u_-, u_+), \\
d_\pi^{(n,\tau)} &= \frac{1}{d_a} \left\| a_{t+\tau}^{(n),e} - \tilde{a}_{\pi,t} \right\|_2^2, \\
C_i &= \min_{\pi \in \Pi} d_\pi^{(n,\tau)}.
\end{aligned} \tag{25}$$

The corridor-return credit in (21) is

$$R_r^{(n,\tau)} = \mathbb{I}[\tau = 0] \mathbb{I}[\rho_t] \hat{\kappa}_t c_{n,s}, \quad \hat{\kappa}_t = \text{clip}(g_r \zeta_t, -1, 1), \tag{26}$$

where $c_{n,s}$ is the steering component of candidate n and ρ_t is the gate of the return layer defined above: the credit is paid once, at the real decision step, and only when the layer may act. It is deliberately model-free. Over a four-step horizon the learned rollout is smooth in its cross-track channel, so candidates whose steering commands point in opposite directions differ by much less in predicted offset than they do in realized offset, and (21) cannot separate a returning command from a lane-keeping one on that channel alone; scoring the command against the measured offset puts the direction back into the ordering. The gap is quantified on the audited draws in Section IV-D. The credit is inactive inside the gate, so it does not bias an ordinary pass, and it is the only term in (21) that is not read from the world model. The active feasibility cost is a soft geometry-dependent penalty:

$$\begin{aligned}
C_f &= \lambda_{\text{geom}} \left\{ \kappa [v - v_{\text{ref},t}]_+ / v_g \right. \\
&\quad + [|\hat{\ell}| - \ell_f]_+ (1 + \beta_\ell \kappa) \\
&\quad + [c_f - \cos(e_\psi)]_+ (1 + \beta_\psi \kappa) \\
&\quad \left. + \beta_g \mathbb{I}[\text{grass}] (1 + \beta_\kappa \kappa) \right\}.
\end{aligned} \tag{27}$$

Here d_a is the action dimension; κ is the absolute wrapped track-angle change divided by the centreline-index lookahead span, not metric curvature. It and the geometry-dependent target speed $v_{\text{ref},t}$ are computed at the real decision step and held fixed during the imagined horizon; v_g is the speed normalization for this penalty. The optional base penalty is disabled in the submitted configuration. This cost is not a control barrier function and does not provide a formal collision or safety guarantee. The quality actor is the active quality-guidance module, and the auxiliary quality head in (14) is an optional training output that is inactive in the submitted checkpoint, where $\lambda_q = 0$.

State feasibility as a separate penalty: Two closed-loop constraints are evaluated on the imagined states and

accumulated as one violation that is priced by λ_ϕ rather than folded into the geometry penalty (27):

$$\begin{aligned}
\phi_g &= \mathbb{I}[\text{back}] + \mathbb{I}[\text{grass}] (1 + [|\hat{\ell}| - \ell_{\text{guard,grass}}]_+ \\
&\quad + [|\hat{\ell}| - \ell_{\text{guard}}]_+ + [c_{\text{guard}} - \cos \hat{e}_\psi]_+, \\
\phi_c &= \sum_j [\ell_{\text{cl}} - |\hat{\rho}_j^\perp|]_+ \mathbb{I}[|\hat{\rho}_j^\perp| \leq \ell_{\text{ab}}] m_j, \\
\phi^{(n,\tau)} &= \phi_g + \phi_c,
\end{aligned} \tag{28}$$

so that J_n in (21) carries the extra term $-\lambda_\phi \sum_{\tau=0}^{H-1} \gamma^\tau \phi^{(n,\tau)}$. In (28), ϕ_g is the guard severity (backward motion, grass, roadside offset, heading), ϕ_c is the swept lateral-clearance shortfall of every longitudinally overlapping neighbour, and m_j is the slot validity mask. ℓ_{cl} , ℓ_{ab} , ℓ_{guard} and the tighter grass threshold $\ell_{\text{guard,grass}}$ are the clearance, alongside-length and guard thresholds, all stated in Table I and expressed in road half-widths, the units of the corrected lateral channel. The stall cost is deliberately not part of this violation: it is the score term C_{live} of (21), priced at λ_{live} , because the guard prices geometry while the stall cost has to trade against progress inside the same sum.

The violation enters the objective as a soft penalty and also orders the pool. Candidates are grouped by $\text{round}(\phi^{(n)}, 10^{-2})$ and the objective only breaks ties inside a group, that is $\tilde{J}_n = J_n - \Lambda \text{round}(\phi^{(n)}, 10^{-2})$ with Λ larger than any objective difference the pool can produce, so the ranking is by $(\phi, -J)$ with feasibility first, the usual order of constrained model-predictive control. Pricing the violation is not by itself sufficient for this pool: an earlier configuration that carried the soft penalty but ordered by the objective alone left the corridor on 43 % of closed-loop steps, while the imitation actor the planner blends towards held $|\hat{\ell}| \leq 0.95$ road half-widths with no grass at all.

The guard is a filter, not a takeover: while $\zeta_t = 1$ its recovery command is one member of $\mathcal{C}_t$ and the remaining candidates are still scored, so the planner keeps control of the committed action. A filter on its own has no fallback, because once the vehicle already sits outside the guard band every candidate inherits a positive violation at the first imagined step; in one archived case the target stayed at $|\hat{\ell}| \approx 5.3$ half-widths and 0 m/s for 2,000 steps, every candidate that would have left the state being scored on an out-of-distribution imagined state. The shield therefore escalates

$$\zeta_t^{\text{esc}} = \mathbb{I} \left[\sum_{s=t-N+1}^t \zeta_s \geq N \right], \tag{29}$$

after which the eligible set $\mathcal{E}_t$ of (17) shrinks to the candidates whose imagined rollout never enters the guarded state, and only when that set is empty does the recovery command replace the committed action. The streak counter $\sum_s \zeta_s$ resets on the first feasible real state, so the escalation is a bounded-duration restriction tied to a measurable condition on the real observations rather than a mode switch on a predicted quantity.

Equations (7)–(29) define the individual operations. Algorithm 1 puts them in the order they run at one physical decision time, and marks where graph refresh ends and imagined propagation begins.

The online loop is $X_t \rightarrow \mathcal{N}_t \rightarrow \bar{O}_t \rightarrow \mathcal{C}_t \rightarrow J_n \rightarrow a_t^e \rightarrow X_{t+1}$, in which $\mathcal{N}_t$ and its slot correspondence are updated

at run time while the packed interface keeps its width. The mechanism is evaluated as part of the complete controller; a selector-only comparison is reported as a diagnostic and does not by itself establish a general completion gain.

IV. EXPERIMENTS AND RESULTS

We evaluate the complete controller against seven references on one shared scenario set: three DLC world-model variants retrained on the same expert data and packed observation layout as the proposed planner, three continuous-control reinforcement-learning agents trained in the same simulator, and a rule-based overtaking expert. The comparison is system-level: the methods share the scenario, observation interface and endpoint definitions but not their training budgets.

A. Experimental setup

Simulator and scenario: Runs use a two-dimensional multi-vehicle racing environment with Box2D contact and its default tyre model, which ships with the released code. It represents neither tyre slip curves nor load transfer nor actuator lag, so the decisions evaluated here are made on planar kinematics and recorded telemetry rather than on a chassis model, and the claims below are at that level of fidelity. Tracks are procedural closed circuits of 263 to 322 tiles of 3.5 m, that is 0.9 to 1.1 km, on a 13.3 m road, so a lap is long enough that an overtake has to be completed rather than merely initiated. A case is a fleet size, a track and a seed. Each of the twelve seeds draws one circuit, and that circuit is reused unchanged at all four fleet sizes (4, 6, 8 and 10 vehicles); no seed is drawn twice, so the twelve circuits are distinct and the 48 cases per method are twelve tracks crossed with four fleet sizes, not 48 independent draws. Within a case every method sees the same track, initial state and opponent behaviour.

The field starts in single file, one vehicle per row and 15 tiles apart, with the evaluated vehicle last so that adjacent

vehicles do not touch. Opponents are driven by the slow-traffic profile, a mix of lane-holding, cruising and yielding rule policies with target speeds between 11 and 15.8 m/s, which leaves an overtake to be earned rather than given. An episode runs for at most 2,200 steps and ends early when the evaluated vehicle completes a lap or when any vehicle leaves the playfield.

Controllers and training budget: Every controller commands steering, throttle and brake at each step, so the interface is identical across methods. The three DLC variants differ only in how the transition model factorizes the field (individual, joint, joint with an observer), and each is trained under our protocol but with its own objective.

The learned methods do not share a budget, and each is reported under the budget it was given rather than rescaled to a common one. Our world model and the three DLC variants are trained for 40,000 gradient steps at batch size 512 on 160 expert episodes, and the DLC actors are regularized toward the recorded expert actions; each variant is retrained over four draws and the proposed controller over five. PPO, SAC and TD3 [53]–[55], in their Stable-Baselines3 implementations [56], are trained for 500,000 environment steps at four vehicles, five training seeds each, and are evaluated beyond their training fleet size on the same 48 cases. Each continuous-control family is therefore reported as a mean over its five seeds together with the range the seeds span, and its paired counts against the proposed controller are formed per seed. These three families were also run at one seed and 50,000 steps; the tenfold budget and the five seeds reported here are what test whether the comparison depended on that setting.

Metrics and the overtake criterion: An overtake is scored from the recorded trajectory, not from a progress counter. Let P denote a physical pass, L sustained lead, C the absence of recorded contact, T on-track execution, S post-pass stability and Q a racing opponent. The tiers are nested,

$$\begin{aligned} E_{\text{pass}} &= P \wedge L \wedge C, \\ E_{\text{valid}} &= E_{\text{pass}} \wedge T, \\ E_{\text{full}} &= E_{\text{valid}} \wedge S, \\ E_{\text{race}} &= E_{\text{full}} \wedge Q. \end{aligned} \tag{30}$$

P requires an approach at a relative along-track distance between -20 and -4 m followed by a crossing of $+4$ m with a 50-step post-pass window available; L requires the relative distance to stay at least 4 m through that window; C excludes recorded ego contacts from approach to the end of the window. T requires the cross-track offset of the evaluated vehicle to stay within one road half-width for at least 70 % of the window [complete -25 , complete $+50$]. S requires finite telemetry, at least 1 m/s, a lateral change of at most 0.10 half-widths per step and a heading within 35° of the local tangent throughout the hold window. Finally, Q requires the overtaken vehicle to still be on the racing surface at the approach sample, defined as offset within 1.5 half-widths, speed at least 4 m/s and not reversed, because driving past a vehicle that has already left the track is not an overtake. E_{race} is the endpoint we report as overtaking success, and Table II reports the physical pass and the strict tier beside it.

Algorithm 1. DNQ-DLC online decision at physical time t .

1: **Input:** $X_t, \mathcal{M}, \mathcal{F}, \pi_b, \pi_q, \Pi_{bg}, \mathcal{A};$ $H, \gamma, N;$ capacities
 $K_{\text{cap}}, K_{\text{model}}, B$
2: **Output:** committed target action a_t^e

3: From the runtime-exposed candidate set in X_t , construct the gated relation set $\mathcal{N}_t$
4: For each $j \in \mathcal{N}_t$, compute its relevance score ρ_t^j using (5)
5: Order and retain represented neighbours $(j_t^1, \dots, j_t^{K_t})$; define $\tilde{\mathcal{N}}_t$ using (7)
6: Retain the first $\min(K_t, K_{\text{model}})$ relations; assign them to slots and mask unused slots
7: Pack $\mathcal{O}_t$ from the slot relations and masks using (8)–(9)
8: Update the feasibility streak and ζ_t^{esc} using (29)
9: Generate $\mathcal{C}_t$ using (16) and budget B , adding the recovery command when $\zeta_t = 1$
10: **for each** $c_n \in \mathcal{C}_t$ **do**
11: Initialize $\tilde{\mathcal{O}}_{t|t}^{(n)} \leftarrow \tilde{\mathcal{O}}_t$, fix $a_t^{(n),e} \leftarrow c_n$, and set $J_n \leftarrow 0$
12: **for** $\tau = 0, \dots, H - 1$ **do**
13: Generate background actions; use (19) for target continuation
14: Predict one step with (20) using fixed transition-tensor slot positions
15: Carry forward $m_t^{i,s}$; quality-actor repacking uses predicted relations only
16: Accumulate the objective J_n and the violation $\phi^{(n)}$ from (21)–(27)
17: **end for**
18: **end for;** order $\mathcal{C}_t$ by $(\phi, -J)$
19: **if** $\zeta_t^{\text{esc}} = 1$ **then** reduce $\mathcal{C}_t$ to the candidates with $\phi^{(n)} = 0$ and commit a_t^e if that set is empty
20: Commit $a_t^e \leftarrow a_t^{(n^*),e}$ from the leading candidate; execute it, observe X_{t+1} , and repeat from the new real observation

TABLE I

FROZEN CONSTANTS OF THE SUBMITTED CONTROLLER: SELECTOR, PLANNER GEOMETRY AND SHIELD. LENGTHS ARE IN METRES, THE GATE AND GUARD DISTANCES IN ROAD HALF WIDTHS, AND c_f , c_{guard} , $c_{\psi, \min}$ AND c_h ARE HEADING COSINES.

Quantity	Symbol	Value	Quantity	Symbol	Value
Rollout horizon	H	4	Progress weight	λ_p	1.20
Candidate budget	B	12	Overtaking weight	λ_o	2.40
Relation slots	K_{model}	3	Grass, lateral, close-gap	$\lambda_g, \lambda_\ell, \lambda_c$	2.00, 0.42, 0.65
Runtime capacity	K_{cap}	3	Quality increments	$\Delta\lambda_o, \Delta\lambda_\ell, \Delta\lambda_g, \Delta\lambda_c$	0.60, 0.60, 1.00, 0.20
Row, relation, packed dim.	d_ϕ, d_r, d_o	17, 7, 41	Risk, unc., live, guard	$\lambda_\chi, \lambda_u, \lambda_{\text{live}}, \lambda_\phi$	1.65, 0.40, 1.50, 5.00
World, actor width	d_h, d_p	192, 128	Clearance, alongside, gate	$\ell_{cl}, \ell_{ab}, g_c, v_c$	3.5, 5.5, 10.0, 18.0
Discount	γ	0.985	Guard lateral, grass, heading	$\ell_{\text{guard}}, \ell_{\text{guard, grass}}, c_{\text{guard}}$	0.76, 0.66, 0.55
Continuation blend	α_q	0.35	Lane, edge, feasibility	ℓ_e, ℓ_g, ℓ_f	0.40, 0.80, 0.40
Reference, live speed	$v_{\text{ref}}, v_{\text{live}}$	22.0, 17.0	Lane, heading costs	$w_\psi, w_{\ell, e}, c_f$	0.55, 1.8, 0.78
Loss weights	$\alpha_\chi, \alpha_r, \alpha_p$	1.20, 0.40, 0.35	Heading gate, weight	$c_{\psi, \min}, w_{\psi, e}$	0.82, 1.2
Auxiliary head, variance scale	$\lambda_q, \mathcal{L}_v$	0, 0.01	Grass, backward, lane	$w_{\text{grass}}, w_{\text{back}}, w_{\ell, g}$	2.5, 1.2, 0.8
Action dimension	d_a	3	Pass, ahead, speed	$w_{\text{pass}}, \ell_{\text{clr}}, w_a, w_v, r_{\pm}$	1.2, 4, 0.35, 0.03, 0.4/0.8
Imitation, geometry	λ_i, λ_f	0.06, 0.75	Curvature gains, clip	$\beta_\ell, \beta_\psi, \beta_\kappa, \beta_g, u_-, u_+$	8, 10, 6, 1.2, -2, 4
Cone length, half-width	$L_c, L_\perp$	65, 18	Relevance weights	w_f, w_s, w_r, w_d, c_h	3.6, 2.8, 1.7, 1.1, 0.25
Grading, speed scale	L_g, v_g	65, 8.0	Relevance weights	$w_{v, \rho}, w_{\psi, \rho}, w_{d, \rho}$	0.25, 0.18, 0.015
Intercept radius	R_{loc}	80	Relevance gates (m)	f_f, l_f, f_s, l_s	70, 24, 14, 18
Escalation streak	N	25	Relevance gates (m)	f_r, l_r, v_r	18, 14, 1.0
Escalation fallback	—	geometric recovery	Front window, scale	f_0, f_w, d_0	22, 55, 12

These thresholds are audit criteria on recorded telemetry, and a case passes a tier only if at least one event satisfies every condition in it; failures and censored records stay in the denominator. Alongside the overtake tiers we report corridor discipline from the same runs. The headline containment measures cover the whole episode and therefore use all 48 cases of every method: the fraction of steps flagged on grass and the mean absolute lateral offset in road half widths. A manoeuvre window exists only where a passing event was found, so its denominator is the passing cases of that method; over it we report the fraction of steps beyond one road half-width from the centreline, the graded companion of the T predicate, and the mean absolute lateral offset over those steps. Binary rates use Wilson 95 % intervals, continuous quantities run-level means with normal-approximation intervals, and per-case paired differences are reported against the strongest reference rather than pooled across protocols.

Three columns of Table II are not tier rates. Rank gain is the number of positions the evaluated vehicle gains between the starting grid and the end of the episode, averaged over cases; speed is its mean speed over the episode; and the pass time is the number of steps from the start of the manoeuvre to the crossing. All three are read from the same recorded telemetry as the tiers.

B. Main comparison across fleet sizes

Table II collects the 48 cases of every method. Across the five draws of its training recipe the proposed controller completes 45.8 of them on average, the rule expert 45 and the joint-transition observer variant 45.5, against 41.0 and 41.5 for the other two DLC variants. The draw ranges separate the proposed controller from the individual-transition variant, whose best draw completes 44 of the 48 cases, but not from the joint-transition or observer variants, whose best draws complete 45 and 46, or from the rule expert. Among the continuous-control families the same column is 35.8 for PPO, 10.2 for SAC and 8.2 for TD3, so completion separates the structured controllers from all three, although one PPO seed reaches 46.

Rank gain is the primary outcome: 4.88 positions per case for the proposed controller against 3.98 for the rule expert and 3.98, 4.19 and 2.49 for the three DLC variants, with the continuous-control families at 0.47, 0.10 and 0.11. Every one of the five draws gains between 4.62 and 5.02 positions per case, so each of them is above every draw of every world-model variant, whose own draws span 3.38 to 4.58, 3.79 to 4.58 and 1.79 to 3.19, and above the rule expert. The scored passing events per case are 4.05 against 4.12 for the rule expert and 3.48, 3.92 and 3.91 for the variants, so the proposed controller converts a comparable number of passes into more positions (Fig. 2a,b).

The strict tier separates them (Fig. 2c). Averaged over the draws, the proposed controller reaches it in 20.2 of the 48 cases, against 17.5, 18.5 and 13.5 for the individual, joint and observer variants, 12 for the rule expert and 6.4, 1.6 and 2.2 for the PPO, SAC and TD3 families. Its draw range, 14 to 26, overlaps all three variants, whose own ranges are 15 to 20, 14 to 23 and 11 to 16, but not the rule expert, and its best draw reaches 26 of the 48 cases, above the best draw of every comparator. The count is insensitive to how the tier is defined: perturbing each of its seven thresholds in both directions over fifteen variants moves the five-draw mean between 18.6 and 20.6 cases against 20.2 as reported, and under every perturbation the best draw of the proposed controller still reaches at least 25 cases, above the 23 of the best comparator.

Because the methods run on the same 48 cases, the informative quantity is the paired count. Against the rule expert the proposed controller wins 13.8 cases and loses 5.6 on average, with per-draw counts of 11–4, 8–6, 16–5, 15–8 and 19–5. Against the three world-model variants, pooled over the four draws of each, the submitted draw, the first of the five, is ahead more often than behind on the strict tier against all three, 10 against 4.5, 9 against 7.5 and 9.5 against 9 cases for the observer, individual and joint variants; on the racing-rival tier it leads the observer variant, 6 against 4, is level with the individual variant at 6.5 against 7, and trails the joint variant, 6.5 against 9. Completion is a tie at this sample size, 45.8 against 45 cases, so the two differ in what happens inside a

TABLE II

CASE-LEVEL OUTCOMES OVER THE 48 SHARED CASES. EVERY LEARNED METHOD IS THE MEAN OVER THE DRAWS OF ITS RECIPE AND THE BRACKETS SPAN THOSE DRAWS; THE TIERS ARE DEFINED IN SECTION IV-A.

Method	P	E_{full}	E_{race}	grass	$ \ell $ (hw)	speed (m/s)	t_{pass} (steps)	rank gain
DNQ-DLC (ours)	45.8 [45, 47]	20.2 [14, 26]	11.0 [6, 15]	0.157 [0.124, 0.199]	1.11 [0.81, 1.38]	15.9	281	4.88
Ours, shield off	44.6 [38, 48]	20.6 [17, 24]	8.8 [3, 13]	0.169 [0.127, 0.249]	0.86 [0.80, 0.98]	15.2	245	4.68
Rule expert	45	12	7	0.299	1.23	17.1	274	3.98
DLC-IT (matched)	41.0 [38, 44]	17.5 [15, 20]	10.5 [9, 12]	0.311 [0.250, 0.372]	1.16 [0.93, 1.39]	19.1	197	3.98
DLC-JT (matched)	41.5 [38, 45]	18.5 [14, 23]	12.5 [9, 16]	0.364 [0.313, 0.415]	1.93 [1.52, 2.34]	18.8	220	4.19
DLC-JTO (matched)	45.5 [45, 46]	13.5 [11, 16]	8.0 [6, 10]	0.457 [0.418, 0.497]	1.69 [1.41, 1.97]	17.7	222	2.49
PPO	35.8 [28, 46]	6.4 [0, 22]	3.2 [0, 16]	0.721	2.81	44.6	63	0.47
SAC	10.2 [0, 15]	1.6 [0, 5]	0	0.331	1.16	21.8	121	0.10
TD3	8.2 [0, 13]	2.2 [0, 5]	0.2 [0, 1]	0.245	0.80	20.8	166	0.11

pass rather than in how many they make. The pairing with the rule expert is not significant in every draw: McNemar’s exact test on those discordant pairs gives $p = 0.12, 0.79, 0.027, 0.21$ and 0.007 over the five draws, so the significance of that comparison belongs to the draw rather than to the recipe, and the endpoint is reported as counts and intervals. On rank gain the paired difference against the rule expert averages $+0.90$ positions per case and is positive in 18 to 20 of the 48 cases, negative in 7 to 13 and exactly zero in the rest. The other comparisons were not pre-specified and there are twelve of them in this section, so they carry no test.

Draws of the world-model training: The transition model is one draw from a stochastic schedule, so five draws were trained from the same recipe on the same data and evaluated on the 48 cases, and every learned row of Table II is the mean over the draws of its recipe. Completion is insensitive to the draw, 45 to 47 of the 48 cases; containment moves from 0.124 to 0.199 of the episode on grass, the strict tier from 14 to 26 cases and rank gain from 4.62 to 5.02 positions per case. On containment and rank gain the draw spread is narrower than the distance to the comparators: the highest grass fraction among the five draws, 0.199, is below the lowest of the twelve world-model draws, 0.250, and the lowest rank gain among them, 4.62, is above the highest of those twelve, 4.58. On the strict tier the spread is wider than the gaps: the range overlaps all three variants, and no pair of learned recipes has disjoint ranges. The three world-model variants are draws of the same kind, and their own retraining moves the strict tier between 15 and 20, 14 and 23 and 11 and 16 cases and the whole-episode grass fraction between 0.250 and 0.372, 0.313 and 0.415 and 0.418 and 0.497, so the claims of this section are about the recipes rather than about individual checkpoints.

What does not separate the draws is prediction error. On episode-disjoint validation, which no checkpoint was selected on, the submitted checkpoint has the largest one-step error of the five on both held-out tracks, 0.049 and 0.037 against 0.025 to 0.042 and 0.022 to 0.030, and it has the largest five-step error there as well, 0.181 and 0.096 against 0.109 to 0.120 and 0.084 to 0.089, as it does on the training distribution, 0.066 against 0.065. What the planner does with the prediction differs as well: the share of steps on which the recovery shield is active ranges from 0.30 to 0.46 across the five draws (0.35 to 0.45 once averaged over the four fleet sizes, as in Table IV), but in this set of draws that share does not track the strict-tier count, which is why the draw is reported as a property of the

recipe rather than explained by a single internal signal.

The gap is not spread evenly over the clauses. The ladder P , the pass tier $E_{\text{pass}} = P \wedge L \wedge C$, E_{valid} , E_{full} , E_{race} runs 45.8, 21.2, 21.0, 20.2, 11.0 for the proposed controller, 45, 14, 13, 12, 7 for the rule expert and 45.5, 25.0, 15.5, 13.5, 8.0 for the observer variant. Given a completed pass, the proposed controller keeps the lead in 68 % of cases and record no contact in 75 %, against 78 and 47 % for the rule expert and 81 and 83 % for the joint-transition variant, whose own completion rate is 41.5 of 48 cases; the on-track and stability clauses hold for 98 and 95 % of our passes against 82 and 71 % for the rule expert. Relative to the rule expert the proposed controller gives up a little lead persistence and gains much more on the remaining clauses, so the gap between 45 and 12 strict-tier cases is not in making the pass but in what survives it.

Containment separates the proposed controller from the comparisons as well, though by a narrower margin than the tier counts imply. Measured over the whole episode, and therefore on all 48 cases of every method, it spends the smallest share of its steps on grass of the nine rows, 15.7 %, against 24.5 % for TD3, 29.9 % for the rule expert, 31.1, 36.4 and 45.7 % for the three DLC variants, 33.1 % for SAC and 72.1 % for PPO. Its mean absolute lateral offset, 1.11 half widths, is the lowest of the four structured controllers and the rule expert, which stand at 1.16, 1.93, 1.69 and 1.23 half widths, but two continuous-control families that make almost no passes sit lower, TD3 at 0.80 and SAC at 1.16, on a scale where the road edge is one half width. What the means do not show is the spread (Fig. 2(d) and (e)): three quarters of our cases stay below 0.17 of the episode on grass and a tenth exceed 0.5, and our lateral interquartile range, 0.56 to 0.80 half widths, is narrower than that of every structured comparator, 0.35 to 1.71 for the rule expert and 0.50 to 1.41, 0.63 to 1.98 and 0.62 to 2.30 for the individual, joint and observer variants; only the two continuous-control families that make almost no passes sit lower.

Over the manoeuvre window the order changes. The window is defined around a completed pass, so a method is measured only on the cases that produce one: 229 of the 240 case-draw pairs of the proposed controller and 45 of 48 for the rule expert, against 179, 51 and 41 of 240 for the PPO, SAC and TD3 families. On those subsets the proposed controller spends 8.4 % of the window beyond one road half-width from the centreline against 35.1 % for the rule expert,

9.3 to 25.5 % for the three world-model variants and 43.2 % for PPO, whose passes are numerous but largely driven off the surface, at a mean lateral offset of 1.87 half widths against 0.70 for the proposed controller, 1.16 for the rule expert and 0.72 to 0.83 for the world-model variants. The two agents that pass least, SAC and TD3, are tighter than the proposed controller inside the few windows they complete, 8.7 % and 4.7 % off the surface at 0.49 and 0.46 half widths; among the structured controllers the proposed controller is the tightest on that measure as well. A clean pass and a frequent pass are separate achievements in this field.

Speed does not rank the methods as the endpoints do: the individual and joint variants drive faster than the proposed controller, 19.1 and 18.8 m/s against 15.9 m/s, and pass sooner, in 197 and 220 steps against 281, yet both complete fewer cases and gain fewer positions (Table II).

C. Fleet size and the structure of the manoeuvre

Completion does not degrade with fleet size for the proposed controller: 57, 55, 59 and 58 of the sixty case-draw pairs at four, six, eight and ten vehicles. The strict tier rises with the field instead, 18, 23, 29 and 31 of the same pairs, and mean rank gain rises from 2.5 to 7.8 positions per case, because a larger field offers both more vehicles to pass and more positions to gain. The racing-rival tier does not follow it, 12, 14, 20 and 9: at ten vehicles the passes that satisfy the strict tier are usually passes of a vehicle that has already left the racing surface, and excluding those is what that tier adds. A pass at ten vehicles also has to be made against nine opponents whose speeds differ by less than 5 m/s, so the approach is longer and the hold window is more often interrupted.

At eight vehicles the approach to a pass takes 125 steps and the alongside phase 111, against 98 and 163 for the rule expert: the rule expert reaches the vehicle sooner and then spends half again as long beside it, and the side-by-side phase is where contact and containment failures originate (Figs. 5 and 3). Counted over all four fleet sizes, it also spends 22 steps off the racing surface per completed pass against 66 for the rule expert, 53 for the observer variant and 16 for the individual-transition variant. The vehicle being passed is not passive: over the same pooled passes it slows by at least 2 m/s inside the manoeuvre window in 8 % of them, against 24 % for the rule expert. Table III carries the failure profile: of the 27.8 cases where the proposed controller misses E_{full} , 17 lose the lead before the window ends and 13.6 record contact, whereas the rule expert loses 27 cases to contact and 16 to stability over the same 48 cases.

Continuous-control families under retraining: The three continuous-control families are a different operating regime rather than a uniformly weaker one, and the retrained numbers sharpen the contrast that the single-seed ones only suggested. PPO completes 35.8 physical passes of 48 on average, fewer than any of the five structured controllers but nearly four times as many as SAC, at a mean speed of 44.6 m/s while spending 72.1 % of the episode on grass; SAC and TD3 complete 10.2 and 8.2 at 21.8 and 20.8 m/s and 33.1 and 24.5 % grass. What they do not convert is the pass: the strict tier falls to 6.4, 1.6

TABLE III
CASES THAT FAIL A CLAUSE OVER THE 48 CASES OF EACH METHOD, AVERAGED OVER THE DRAWS OF ITS RECIPE. COLUMNS ARE NOT EXCLUSIVE; A CASE WITH NO AUDITABLE PASS COUNTS UNDER EVERY COLUMN.

Method	no lead	contact	off track	stability	non-racing rival
DNQ-DLC (ours)	17	13.6	3	4.6	5.6
Rule expert	13	27	11	16	4
DLC-IT	19.5	16.0	11.5	13.5	12.0
DLC-JT	14.5	13.5	8.5	8.0	11.0
DLC-JTO	15.0	14.5	8.0	5.0	12.0

and 2.2 cases against the 20.2 of the proposed controller, and the racing-rival tier to 3.2, 0.0 and 0.2 against 11.0.

Spread across training seeds: The fifteen retrained agents disagree widely with one another, in both directions. The best of them is a PPO seed that reaches the strict tier in 22 cases and the racing-rival tier in 16, above the proposed controller's five-draw mean of 20.2 and 11.0 and, on the racing-rival tier, above its whole 6-to-15 range, while spending 73.4 % of the episode on grass at a mean lateral error of 2.70 half widths against 15.7 % and 1.11 for the proposed controller. The best-contained of the fifteen is a TD3 seed at 2.7 % grass that completes no pass at all. Either the endpoint count or the containment column can therefore be attained without the other, which is why both are reported and why the paired counts of Section IV-B are read against Fig. 2f.

D. Mechanism evidence from the same runs

The three claims of Section III are measured from the runs that produced Table II, because the planner records its admission decision, its imagined rollout and its candidate scores at every step. Only one of the three is tested by removing the component: the admission rule is replaced by two alternatives below. The imagined rollout and the quality score are not, so for those two the evidence is the measured accuracy of the rollout and the measured departure from the rule anchor, which show that both are active in the executed decisions rather than that either is what produces the endpoint.

Size and composition of the interaction set: Exposure grows linearly with the field while the admitted set saturates at the slot budget of three: three, five, seven and nine opposing vehicles are exposed at four, six, eight and ten vehicles, against 1.85, 2.30, 2.66 and 2.85 admitted. The admitted set is part of the observation the transition model is trained on, not a filter on its input, and its composition changes: pruning fires on 46 % of steps at four vehicles and on every step from six on, so the exposed set is a strict superset of the admitted one in a dense field, and the budget is filled on 54 % of steps at four vehicles against 91 % at ten (Fig. 6a,b).

Choice of the admission rule: Whether the ranking drives the effect was tested by re-packing the released expert episodes offline under three rules at the same budget of three slots: the interaction ranking, the three nearest opponents, and the first three opponents in identity order. The three datasets therefore share the episode plan, the expert actions, the rewards and the risks exactly and differ only in the packed slots (maximum slot difference 2.0, behaviour difference 0.0 over

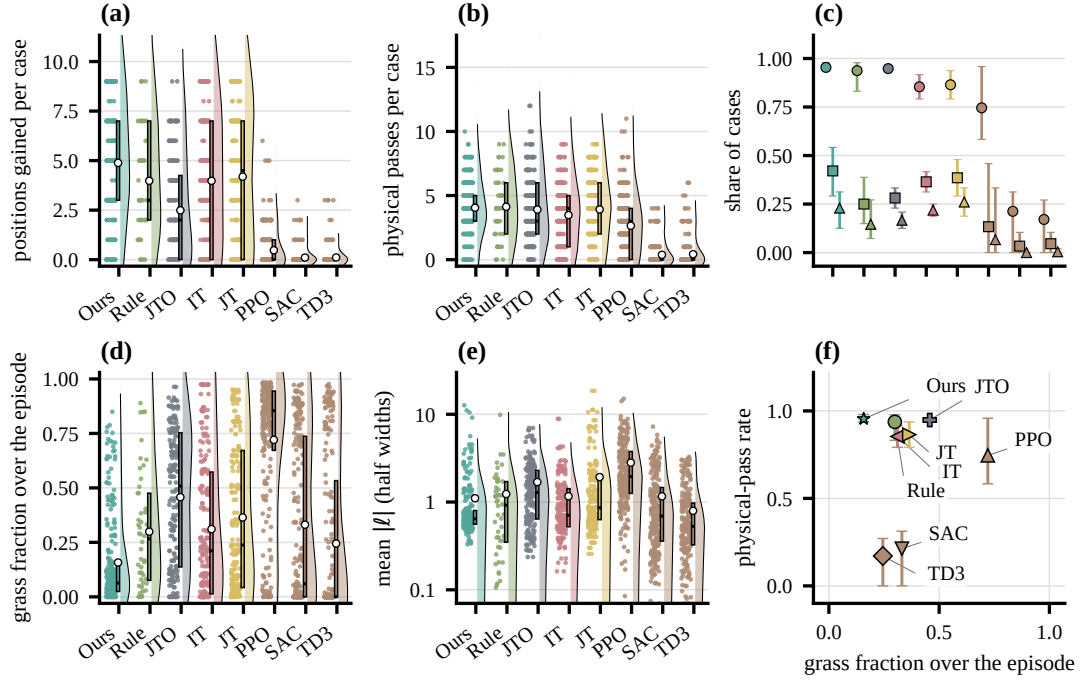


Fig. 2. Main comparison over the 48 shared cases. (a) Positions gained per case. (b) Scored passing events per case. (c) Share of cases reaching P (circles), E_{full} (squares) and E_{race} (triangles), with Wilson 95 % intervals. (d) Grass fraction over the whole episode. (e) Mean absolute lateral offset in road half widths. (f) Physical-pass rate against grass fraction. Panels (a), (b), (d), (e): rainclouds (dots, box, density).

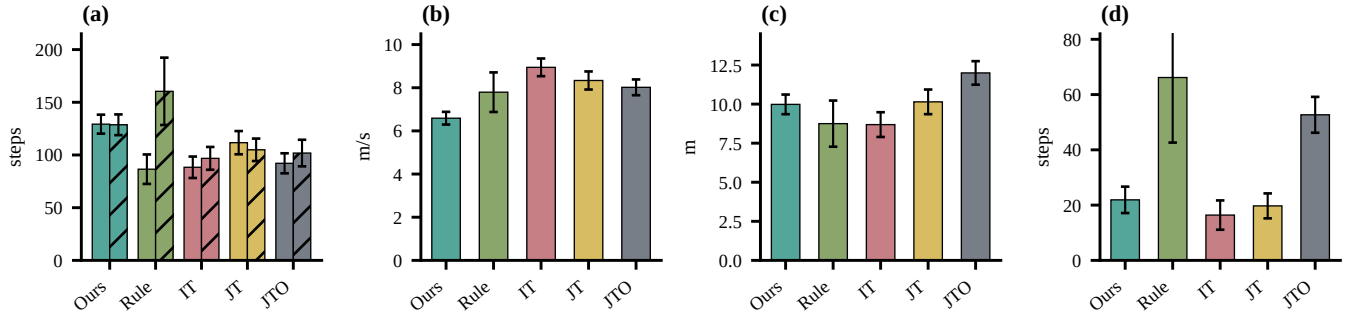


Fig. 3. Completed passes of the five structured controllers, pooled over the four fleet sizes; means with the 95 % interval of the mean. (a) Approach (solid) and alongside (hatched) phase. (b) Speed advantage while alongside. (c) Closest centre-to-centre distance to the rival. (d) Steps off the racing surface.

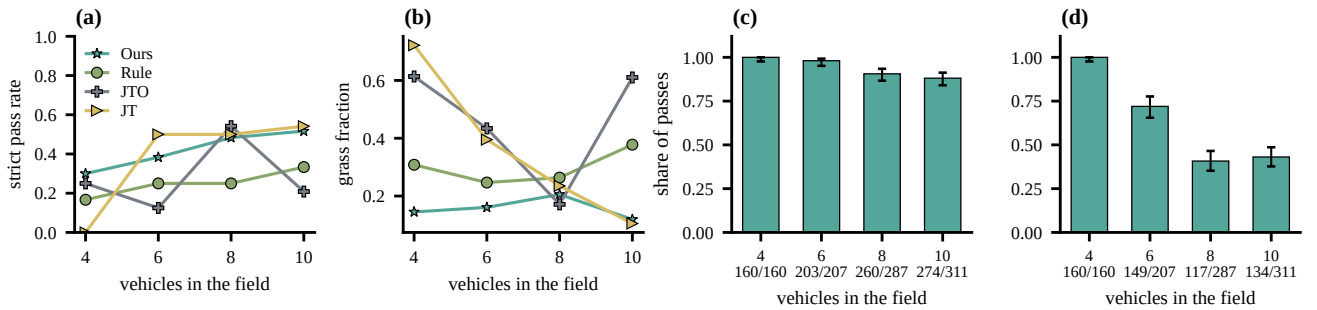


Fig. 4. Fleet-size dependence of the four methods that complete most cases (a, b) and membership of the overtaken vehicle under the proposed controller (c, d). (a) Strict-tier rate. (b) Grass fraction over the whole episode. (c) Share of passes whose partner entered the admitted set during the window. (d) Share whose partner was still admitted at completion. Bars carry Wilson 95 % intervals and the raw counts.

352000 transitions). Retrained under the identical recipe with a single-member model, and evaluated on the same 48 cases, the

interaction ranking reaches the strict tier in 21 cases against 17 for the nearest rule and 15 for the fixed identity window, and

TABLE IV

INTERACTION-GRAPH AND PLANNER STATISTICS OF THE PROPOSED CONTROLLER, MEANS OVER THE SIXTY CASES AT EACH FLEET SIZE. $|\mathcal{N}_t|$ AND K_t ARE THE EXPOSED AND ADMITTED NEIGHBOUR COUNTS, π AND χ THE SHARES OF STEPS THAT PRUNE AN EXPOSED NEIGHBOUR AND CHANGE THE ADMITTED SET, AND D THE DISTINCT ADMITTED NEIGHBOURS PER EPISODE. $\varepsilon_1, \varepsilon_4$ ARE THE IMAGINED EGO-POSITION ERRORS IN METRES AT HORIZONS ONE AND FOUR; ρ_p, ρ_s, ρ_o THE SHARES OF DECISION STEPS WITH A FEASIBLE CANDIDATE, WITH THE SHIELD ACTIVE, AND LEAVING THE RULE ANCHOR.

Fleet size	$ \mathcal{N}_t $	K_t	π	χ	D	ε_1	ε_4	ρ_p	ρ_s	ρ_o
4	3.00	1.85	0.46	0.00	3.0	0.45	0.85	0.71	0.37	0.92
6	5.00	2.30	1.00	0.02	4.7	0.46	0.98	0.67	0.42	0.92
8	7.00	2.66	1.00	0.05	6.7	0.47	1.18	0.63	0.45	0.93
10	9.00	2.85	1.00	0.06	8.3	0.46	1.05	0.74	0.35	0.92

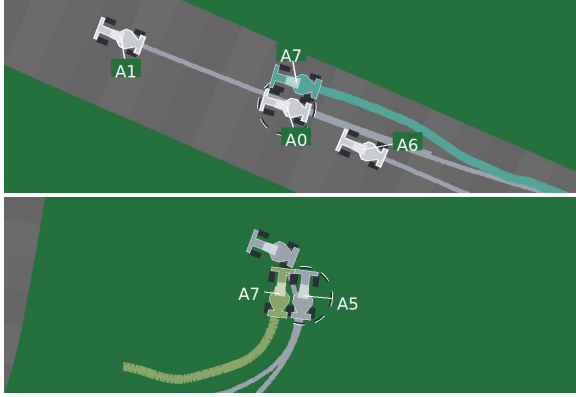


Fig. 5. Closest approach to a rival in one eight-vehicle case, episode seed 29: the proposed controller at step 946 (top) and the rule expert at step 459 (bottom). The proposed controller admits rivals 0, 1 and 6 at its step. The evaluated car A7 is drawn in its controller's colour, every car carries its field index, and admitted rivals are in a lighter tint with a thicker outline.

the racing-rival tier in 14, 7 and 9. The rank gain falls from 4.96 to 4.52 and 2.73 positions per case, the whole-episode grass fraction rises from 0.179 to 0.238 and 0.376, and the mean lateral error from 0.79 to 0.97 and 2.42 half widths. A window that is never re-selected is also behind on the physical pass P , 36 of 48 against 46, a paired difference of -10 with a bootstrap interval excluding zero. Proximity alone carries the pass, which the nearest rule shows by matching 46 of 48 and leading on E_{pass} , 27 against 24, but not the later conditions: it loses more of those passes to the on-track and stability clauses, so the strict tier falls to 17 against 21 and the racing-rival tier to 7 against 14. At that tier the two rules disagree in 13 cases, and the nearest rule is ahead in 3 of them and behind in 10.

Membership of the overtaken vehicle: Reading the same runs by event rather than by case makes set membership measurable: the five draws together yield 965 auditable passing events, and the partner is inside the admitted set somewhere in the window in 897 of them, 93.0% (Wilson [91, 94]): 160 of 160 at four vehicles, 203 of 207 at six, 260 of 287 at eight and 274 of 311 at ten (Fig. 4c). The composition also changes, on average 3.5 times within the window (4.9 at eight vehicles, 4.4 at ten), against 0.006 at four. Once the ego has crossed, the partner is behind it and no longer closing, so the selector releases it: it is still admitted at completion in 41% of eight-vehicle and 43% of ten-vehicle passes, against 100% at four (Fig. 4d). Holding it beyond the crossing would spend a slot on a vehicle that is already behind the evaluated one, so the evaluated configuration releases it.

Accuracy of the imagined rollout: The planner scores candidates on a four-step horizon and the imagined ego position is 0.45 to 0.47 m from the realized trajectory after one step and 0.85 to 1.18 m after four, with mean speed errors of 0.73 to 0.85 m/s at one step and 1.59 to 1.98 m/s at four. The four-step error is about a quarter of a vehicle length over the 0.08 s the four-step ranking spans.

Quality guidance and the rule anchor: The planner carries a rule-based anchor and departs from it on 92–93% of decision steps, by a score margin that averages 134 to 237 units, so the quality-guided ranking rather than the anchor sets the executed action almost everywhere. The anchor is c_{anchor} of (16), a rule-based telemetry gate of the same family as the rule-expert reference, so the ranking overrides that expert's own command on almost every step rather than replaying it.

Residual contribution of the recovery shield: The shield is active on 35–45% of steps, the planner has a feasible candidate on 63–74%, the guard's own trigger fires on 13–15%, and on 26–37% of steps the escalated pool holds no feasible manoeuvre, so the geometric recovery controller drives. Disabling the shield over the same five draws leaves the strict tier where it is, 20.6 cases against 20.2, and the containment lead is intact: the whole-episode grass fraction rises from 0.157 to 0.169 but stays below every draw of every comparator, whereas completion falls from 45.8 to 44.6 cases and the racing-rival tier from 11.0 to 8.8 (Table II). The shield therefore secures the last of the clause of the graded endpoint, that the vehicle being passed is still racing, together with two completions, rather than the strict tier itself.

E. Failure cases and boundaries

In two of the 48 runs of the submitted draw of the proposed controller, at six vehicles seed 32 and at eight vehicles seed 22, the evaluated vehicle covers 0.16 and 0.23 of a lap with no auditable passing event, confined behind a slower vehicle early and never recovering the speed to open a gap. Both count as failures in every rate of Table II, but they do not limit the strict rate at ten vehicles, where every case completes and the limit is the hold window, interrupted more often as the field grows.

Two boundaries limit what the tables support. The strict tier is an audit criterion, not a validated safety limit, and the references are not trained to convergence, so the comparison characterizes the systems as evaluated rather than ranking algorithms under optimal training. Nothing here is a claim about a physical vehicle.

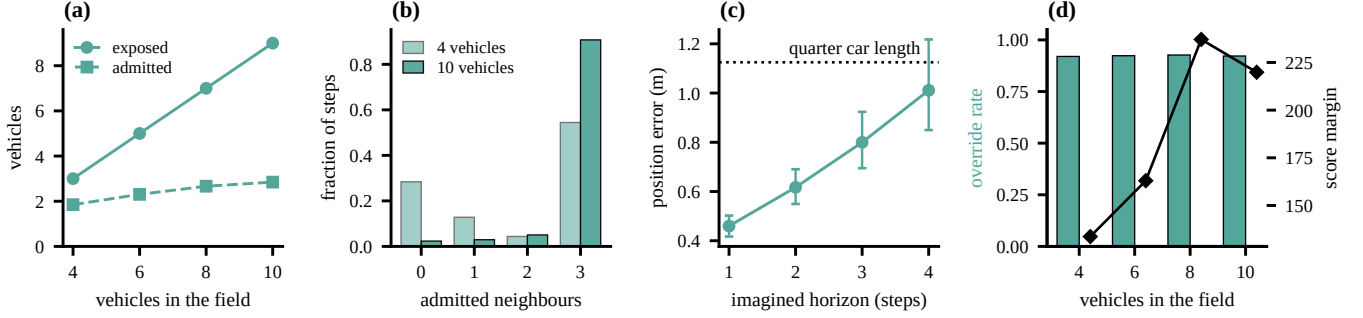


Fig. 6. Controller-internal signals of the proposed controller, from the same runs as Table II. (a) Exposed and admitted neighbours against field size. (b) Admitted count at the smallest and largest fields. (c) Imagined position error against horizon. (d) Override rate (bars) and mean score margin (line).

V. CONCLUSION AND FUTURE WORK

This paper addressed closed-loop overtaking in multi-vehicle autonomous racing, where the opponent that constrains a pass can change while the pass is under way. DNQ-DLC selects the currently relevant opponents from each real observation, predicts their short-horizon interaction with candidate ego actions through an action-conditioned graph world model, and ranks candidates by progress together with containment, interaction quality, boundary risk and uncertainty before committing the first action.

The evaluation compared the complete controller with rule-based and learned references over 48 shared cases per method. Averaged over the five draws of its recipe the controller completes 45.8 of 48 cases and reaches the strict tier in 20.2 of them, against 12 for the rule expert and 13.5 to 18.5 for the three world-model variants. What separates it from the rule expert is not the number of passes, which the expert matches, but the fraction of those passes that survive the containment and stability conditions. The result is bounded in three ways. The shield remains active on 30 to 46 % of steps, and removing it leaves the strict tier unchanged while costing the racing-rival tier and two completions, so the endpoint that depends on it is the last clause of the graded endpoint rather than the strict tier. The continuous-control references bound what PPO, SAC and TD3 reach at a five-hundred-thousand-step budget under five seeds each; the best of the fifteen seeds passes the proposed controller on the strict tier while spending 73.4 % of the episode on grass. And the strict-tier count of the five draws spans 14 to 26 cases, so the size of the margin over the rule expert belongs to the draw, which is why that endpoint is reported as counts and intervals. Every learned row of Table II is the mean over the draws of its recipe, so no column depends on the choice of draw. Future work will keep the representation slots aligned with vehicle identity as the ranking changes, drive the shield activation toward zero, and move the planner to closed-course vehicle experiments.

DATA AND CODE AVAILABILITY

The simulator, controller, evaluation and analysis code, the checkpoints, the configurations and the per-case evaluation and audit tables behind Section IV are at <https://github.com/chocho2222/DNQ-DLC>, released as the content-addressed snapshot *tits-2026-09-29* (1755 files, 97.1 MB); the recorded

trajectories are 8.5 GB for the retrained baselines alone and are not part of it. Every number in Section IV is computed from those tables. The snapshot adds, relative to the earlier one, the fifteen strict continuous-control checkpoints with their tier and containment audits, the further transition-model draws, the five-draw and fused-ensemble runs, the held-out prediction and recovery-deferral checks, the re-run of the admission-rule control under the corrected planner, and the tables behind Figs. 3 and 5. The digest, SHA-256 over sorted $\langle \text{root} \rangle / \langle \text{path} \rangle : \langle \text{digest} \rangle$ lines, is `2d3af4413d04ff96bfd1a85bdc8c9d1b57f48f2a40b3cdac46dd8913225ece98`. Recomputing it over a local copy confirms the copy is the evaluated one; the audit roots are listed in the released manifest.

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